



Smart Robot For Face Recognition

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This article describes how you can design a smart robot that can recognise your face and of other regular visitors. If the robot recognises correctly, it will greet and call out the name.

The project can be used for security purposes through live streaming video using a camera along with this

system. The author's prototype being used for testing is shown in Fig. 1.

You can use any face mask, including the Tahta robot mask available in the market. This project uses an InMoov robot open source face mask, as shown in Fig. 2.

The components required for this project are listed in Table 1.

Assembling the robot face

Assembling steps may vary depending on the shape and size of the robot's head. For testing, we used the InMoov robot head created using a 3D printer. You can even 3D print your own face and use it as a robot head, or get a 3D-printed robot head from thingiverse.com.

Two OLED display modules (DIS1 and DIS2) are used as the robot's eyes. Solder both the display modules and make proper connections. Make sure to securely screw the OLED displays onto the eyepiece of the robot head, as shown in Fig. 3. OLED connections with Arduino are listed in Table 2.



Fig. 1: Author's prototype during testing

Components Required for the Project Table 1	
Components	Quantity
Servo motors (5V)	2
OLED SSD1306s (DIS1 AND DIS2)	2
Arduino Nano	1
Raspberry Pi 4	1
Raspberry Pi camera	1
Amplifier speaker	1
Thin metallic rods (7cm long)	3
A hard-cardboard or wooden base to mount the robot head	1
USB data cable for Arduino	1
5V power adaptor	1
Connecting wires	Red and black hook-up wires
Robot face mask (InMoov or similar)	1



Fig. 2: InMoov face mask

OLED Connections with Arduino Table 2	
Arduino Nano	OLED
GND	GND
5V	VCC
A4	SDA
A5	SCL